

NG LIH SHENG

Full Stack Software Developer | LLM Applications | Agentic AI | React, TypeScript, Laravel, AWS

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PROFESSIONAL SUMMARY

Full Stack Software Developer with 5+ years of experience in enterprise web, HR, analytics, and automation systems. Builds full-stack and AI-enabled products with React, TypeScript, Angular, Laravel, PostgreSQL, and AWS. Experience includes multi-tenant BI, agentic AI, RAG, Row-Level Security, structured LLM outputs, and Langfuse.

CORE SKILLS

Frontend: React.js, Next.js, TypeScript, JavaScript, Angular, Mantine, React Leaflet, dnd-kit, Tailwind CSS

Backend & Data: PHP, Laravel, REST APIs, Microservices, MySQL, PostgreSQL, Prisma, Supabase, Row-Level Security, Multi-Tenant Architecture

Cloud & DevOps: AWS EC2, S3, Lambda, SQS, Docker, GitHub Actions, CI/CD, Supabase Edge Functions, Google Cloud, Cloudflare, systemd

Testing & Quality: Cypress, Jest, Vitest, PHPUnit, Postman, code review, test automation, debugging, production support

AI / LLM / Agents: LLM applications, agentic AI, agent orchestration, RAG workflows, structured LLM outputs, prompt engineering, Google ADK, Langfuse, Qwen/LoRA, Ollama/LM Studio, OpenAB

PROFESSIONAL EXPERIENCE

Software Developer (Full Stack) | Herd HR - Singapore

Jul 2025 - Present

Tech: Laravel, Angular, PostgreSQL, Apache ECharts, AWS Cognito, Python, Microservices

- Maintain and enhance a Laravel + Angular HR platform, including workforce, roster, leave, reporting, reminders, and background jobs.
- Refactor legacy modules and resolve production issues to improve system reliability and maintainability.
- Appointed to design and build WizHR, a multi-tenant HR analytics and BI platform for multiple companies and countries.
- Design WizHR with Angular, Laravel, PostgreSQL, and reusable Apache ECharts components, supporting configurable dashboards and HR data imports.
- Design tenant and data isolation with PostgreSQL Row-Level Security and fine-grained access control.
- Design AI-assisted analytics through a Python service that uses authorized Laravel APIs and preserves user access rules.

Software Developer (Full Stack) | AYP Group - Johor Bahru

Feb 2024 - Jun 2025

Tech: TypeScript, Next.js, Laravel, AWS, Cypress, Jest, MySQL, CI/CD

- Built and maintained full-stack applications with TypeScript, Next.js, Laravel, and AWS for APAC business users.
- Wrote Jest and Cypress tests and maintained over 85% test coverage for key application areas.
- Automated staging and production CI/CD pipelines to reduce manual deployment work and release risk.
- Reviewed code and joined architecture and sprint discussions to improve delivery quality and scalability.

Senior Software Developer | XeerSoft Sdn Bhd - Johor Bahru

Jan 2021 - Feb 2024

Tech: React.js, PHP, JavaScript, REST APIs, MVC

- Led junior developers and interns in daily development, code quality, and feature delivery.
- Redesigned major frontend modules with React.js and MVC to improve usability and maintainability.
- Built reusable React components and custom hooks to reduce duplicate code.
- Integrated REST APIs and resolved cross-origin, latency, and data synchronization issues.

SELECTED PROJECTS

GeminiAgent - Agentic Tuition Centre Operations Platform | Personal Project

Tech: Python, Google ADK, React, Firebase, Google Drive/Sheets APIs, Telegram, Langfuse, Google Cloud

- Built an agentic tuition-centre platform with AI workflows for leads, students, classes, appointments, materials, and admin tasks.
- Integrated Google Drive, Sheets, Telegram, Google authentication, and a web admin portal with conversational agent access.
- Added tracing and evaluation to measure agent behavior, prompt changes, and workflow quality.

LinkUp - AI-Assisted Schema Matching Platform | Personal Project

Tech: Next.js, React, TypeScript, Prisma, PostgreSQL, ExcelJS, Zod, Langfuse, Ollama/LM Studio

- Built a four-step workspace that maps Excel and CSV data to target JSON schemas with AI-assisted review and validation.
- Combined exact matching, synonyms, heuristics, LLM suggestions, confidence scores, duplicate checks, and transform rules.
- Added workbook profiling, data preview, reusable templates, draft persistence, JSON Schema validation, and Langfuse tracing.

Ops-Room - OpenAB Multi-Agent Control Surface | Personal Project

Tech: Node.js, OpenAB, GitHub Webhooks, Docker, Cloudflare Tunnel, systemd

- Built an operations control surface to configure, launch, and monitor multiple OpenAB-backed agents.
- Added GitHub webhook and polling flows to route tasks, create workspaces, track state, and support issue and PR workflows.
- Built a read-only dashboard for agent status, container health, GitHub polling, restart counts, logs, and runtime paths.

Trip Planner - AI-Assisted Collaborative Travel Planning Platform | Personal Project

Tech: React, TypeScript, Mantine, React Leaflet, dnd-kit, Supabase, PostgreSQL, Edge Functions, RLS

- Built a map-first trip planner with place search, drag-and-drop scheduling, route views, Today Mode, and expense tracking.
- Added passwordless login, cloud persistence, collaborator access, public sharing, and PostgreSQL Row-Level Security.
- Built an AI-assisted import flow that converts travel notes and map links into reviewable trip suggestions.

EDUCATION

Bachelor's Degree in Software Engineering | Universiti Tun Hussein Onn Malaysia

Aug 2017 - Dec 2020 | First Class Honours, CGPA 3.83 / 4.00 | Dean's List for all semesters.

- Chief Secretary, Festival Tiong Hua UTHM 2019. Treasurer, Tiong Hua Chinese Cultural Carnival UTHM.

Malaysian Higher School Certificate (STPM) | English College Johor Bahru

Jun 2015 - Dec 2016 | Science Stream, CGPA 3.08 / 4.00 | President of Bowling Club.